

Lab 04 – Exploratory Data Analysis (EDA) Reflection Journal – Antonio Garcia

Exploratory Data Analysis (EDA) is one of the most important steps in any data science workflow, and this lab helped me understand why. Before building any model, it's essential to understand the structure, patterns, and relationships inside the dataset. In this lab, I worked with the Titanic dataset, which contains passenger information such as age, gender, class, fare, and survival status. Running the provided cells and creating my own visualizations helped me see how EDA reveals insights that aren't obvious from raw numbers alone.

One of the first things I noticed was how many missing values the dataset had, especially in the *Age* and *Cabin* columns. Seeing this in the `.info()` output made it clear why data cleaning is necessary before any modeling. The descriptive statistics also showed wide ranges in fare prices and age, which hinted at the diversity of passengers on the ship.

The visualizations were the most eye-opening part of the lab. The survival rate differences between men and women were immediately obvious in the countplots — women had a much higher chance of surviving. The fare vs. survival boxplot showed that passengers who paid higher fares tended to survive more often, which connects directly to class differences. Higher-class passengers had better access to lifeboats and safer areas of the ship. The port of embarkation experiment also showed interesting patterns, especially how passengers from Southampton (S) had lower survival rates, likely because many of them were third-class travelers.

What surprised me most was how clearly the plots revealed relationships that summary statistics alone couldn't show. A mean or count doesn't tell you how values are distributed, whether there are outliers, or how two variables interact. Seeing the distributions visually made the dataset feel more "alive," and it helped me understand why EDA is considered a detective-style process.

This lab also helped me get more comfortable with Seaborn and Matplotlib. I learned how to fix warnings, adjust plot settings, and interpret visual patterns. These skills will be important in future labs, especially when I start building machine learning models. EDA gives me the foundation to make better decisions about feature selection, cleaning, and preprocessing.